

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286266

Kind Code

A1

Publication Date

September 11, 2025

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SPEAKER MODULE, ANTENNA MODULE AND ELECTRONIC DEVICE

Abstract

A speaker module includes a cavity body, a speaker element and a first antenna. The cavity body includes an outer surface, an inner surface and an accommodation space. The inner surface forms the accommodation space. The speaker element is disposed in the accommodation space. The first antenna is disposed on and extended along the outer surface.

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Family ID: 1000008478677

Appl. No.: 19/070601

Filed: March 05, 2025

Foreign Application Priority Data

TW

113108884

Mar. 11, 2024

Publication Classification

Int. Cl.: H01Q1/22 (20060101); H01Q9/04 (20060101); H01Q9/30 (20060101)

U.S. Cl.:

CPC H01Q1/2266 (20130101); H01Q9/0421 (20130101); H01Q9/30 (20130101);

Background/Summary

RELATED APPLICATIONS

[0001] This application claims the benefit of priority to Taiwan Application Number 113108884, filed on Mar. 11, 2024. The entire content of the above identified application is incorporated herein by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a speaker module, an antenna module and an electronic device, and more particularly, to a multifunction speaker module, a multifunction antenna module, and an electronic device having the speaker module and the antenna module.

Description of Related Art

[0003] With the development trend of modern communication systems, the installation space for antenna modules in electronic devices is usually compressed, especially when the electronic device needs to meet the characteristics of multiple communication systems, multiple antennas, and multiple frequency bands. Therefore, the design of the size and position of the clearance area of the antenna module faces severe challenges.

[0004] In view of the above, there is a need to develop an antenna module and an electronic device that meet both the RF (Radio Frequency) characteristics requirements of communication systems and the space utilization rate of antenna installation.

SUMMARY

[0005] In one aspect, the present disclosure provides a speaker module that includes a cavity body, a speaker element, and a first antenna. The cavity body includes an outer surface, an inner surface, and an accommodation space, and the inner surface forms the accommodation space. The speaker element is disposed in the accommodation space. The first antenna is disposed on and extends along the outer surface.

[0006] In another aspect, the present disclosure provides an electronic device that includes the aforementioned speaker module.

[0007] In yet another aspect, the present disclosure provides an antenna module that includes a metal element and a first antenna. The metal element has a metal surface. When observed along a first direction, a projection of a first branch of the first antenna partially overlaps with a projection of the metal element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

[0009] FIG. 1A is a schematic diagram of an antenna module according to a first embodiment of the present disclosure.

[0010] FIG. 1B is a sectional view along line 1B-1B in FIG. 1A.

[0011] FIG. 1C is an isolation schematic diagram of the antenna module in FIG. 1A.

[0012] FIG. 2 is a schematic diagram of a speaker module according to a second embodiment of the present disclosure.

[0013] FIG. 3 is a schematic diagram of a speaker module according to a third embodiment of the present disclosure.

[0014] FIG. 4 is a schematic diagram of an electronic device according to a fourth embodiment of the present disclosure.

DETAILED DESCRIPTION

[0015] The present disclosure is more particularly described in the following examples that are intended as illustrative only since numerous modifications and variations therein will be apparent

to those skilled in the art. Like numbers in the drawings indicate like components throughout the views. As used in the description herein and throughout the claims that follow, unless the context clearly dictates otherwise, the meaning of “a”, “an”, and “the” includes plural reference, and the meaning of “in” includes “in” and “on”. Titles or subtitles can be used herein for the convenience of a reader, which shall have no influence on the scope of the present disclosure.

[0016] The terms used herein generally have their ordinary meanings in the art. In the case of conflict, the present document, including any definitions given herein, will prevail. The same thing can be expressed in more than one way. Alternative language and synonyms can be used for any term(s) discussed herein, and no special significance is to be placed upon whether a term is elaborated or discussed herein. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms is illustrative only, and in no way limits the scope and meaning of the present disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given herein. Numbering terms such as “first”, “second” or “third” can be used to describe various components, signals or the like, which are for distinguishing one component/signal from another one only, and are not intended to, nor should be construed to impose any substantive limitations on the components, signals or the like.

[0017] FIG. 1A is a schematic diagram of an antenna module **100** according to a first embodiment of the present disclosure. Referring to FIG. 1A, the antenna module **100** includes a metal element and a first antenna **130**. The metal element has an exposed outer surface, a non-exposed inner surface, and an interposed surface, and at least one of which includes a metal (conductive) surface. The metal surface can have a floating ground characteristic or be electrically connected to a grounding element of the electronic device. The metal element can be an electronic component with circuit functions or a metal material element without circuit functions. For example, the metal element can be a speaker (transducer) element **120**, and at least part of the outer surface of the speaker element **120** includes a metal surface, but the present disclosure is not limited thereto. The first antenna **130** includes a first branch **131**. When observed along a first direction x, the projection of the first branch **131** partially overlaps with the projection of the speaker element **120**. This helps the first antenna **130** balance RF characteristics requirements and isolate nearby circuit elements (such as the second antenna **150**, but not limited thereto), thereby reducing interference between the first antenna **130** and nearby circuit elements.

[0018] Specifically, the antenna module **100** can further include a grounding element **160**. One end **132** of the first branch **131** is provided with a feeding point **140** or connected to the grounding element **160**. In this embodiment, one end **132** of the first branch **131** is provided with the feeding point **140**. When observed along the first direction x, the projection of the one end **132** of the first branch **131** does not overlap with the projection of the speaker element **120**. Particularly, the projection of the first branch **131** extends beyond the projection of the speaker element **120**, placing the projections of both ends (the one end **132** and the other end) of the first branch **131** outside the projection range of the speaker element **120**. Thus, when the first antenna according to the present disclosure is a PIFA (Planar Inverted-F Antenna) or a monopole antenna, the feeding point avoids the metal element or speaker element, which helps meet the requirements for RF characteristics and improves isolation.

[0019] FIG. 1B is a sectional view along line 1B-1B in FIG. 1A. Referring to FIG. 1A and FIG. 1B, the first antenna **130** can further include a second branch **142**. The other end of the first branch **131** can be connected to the second branch **142**. Both ends of the second branch **142** are open ends, and the second branch **142** can extend on two non-coplanar sub-surfaces **112**, **113** of the outer surface **111**.

[0020] The first antenna **130** can further include a grounding branch **149**. One end of the grounding branch **149** is connected to the grounding element **160**. When observed along the first direction x, the projection of the grounding branch **149** partially overlaps with the projection of the speaker

element **120** (i.e., the projection of the grounding branch **149** and the projection of the speaker element **120** partially overlap). Thus, the second branch **142** and the grounding branch **149**, in conjunction with the first branch **131** provided with the feeding point **140**, make the first antenna **130** suitable for a design that enhances isolation.

[0021] Furthermore, referring to FIG. 1A and FIG. 1B, when observed along a second direction y, the projection of the second branch **142** can partially overlap with the projection of the speaker element **120**. In other antenna modules according to the present disclosure (not shown), when observed along the second direction, the projection of the second branch partially overlaps with the projection of the speaker element, and when observed along the first direction, the projection of the first branch does not overlap with the projection of the speaker element.

[0022] Referring to FIG. 1A, the antenna module **100** can further include a second antenna **150**. Each of the first antenna **130** and the second antenna **150** extends along the second direction y and a third direction z, i.e., each of the first antenna **130** and the second antenna **150** extends along a plane with the normal direction parallel to the first direction x. The first direction x, the second direction y, and the third direction z are perpendicular to one another. The first antenna **130**, the speaker element **120**, and the second antenna **150** are sequentially arranged along the third direction z. Furthermore, the operating frequency band of the first antenna **130** and the operating frequency band of the second antenna **150** can at least partially overlap. Thus, by arranging a metal element (such as the speaker element **120**) with a metal surface at an appropriate position between the first antenna **130** and the second antenna **150**, interference between the first antenna **130** and the second antenna **150** can be reduced. For example, the first antenna **130** can be a WLAN antenna with an operating frequency of 2.4 GHz to 8 GHz, and the second antenna **150** can be an LTE antenna with an operating frequency of 617 MHz to 6 GHz.

[0023] Moreover, the antenna module **100** of the first embodiment can also be referred to as a speaker (transducer) module. The antenna module **100** includes a cavity body **110**, the speaker (transducer) element **120**, and the first antenna **130**. The cavity body **110** includes an outer surface **111**, an inner surface **115**, and an accommodation space **116**. The accommodation space **116** is formed by the inner surface **115**, and the speaker element **120** is disposed in the accommodation space **116**. The first antenna **130** is disposed on and extends along the outer surface **111**. Thus, the first antenna **130** is designed on the outer surface **111** of the cavity body **110** of the speaker element **120**, achieving the effects of saving space and providing good antenna characteristics simultaneously.

[0024] In specific, the cavity body **110** can be made of a dielectric material (such as plastic, but not limited thereto), and each of the first antenna **130** and the second antenna **150** is made of a conductive (electrically conductive) material. As such, the first antenna **130** and the second antenna **150** can be further manufactured on the cavity body **110** by Laser Direct Structuring (LDS) to enhance the application flexibility of the antenna module **100**.

[0025] The second antenna **150** can be disposed on and extends along the outer surface **111**, and the grounding element **160** can be connected to the outer surface **111**. Thus, by arranging the speaker element **120** with a metal surface at an appropriate position between the first antenna **130** and the second antenna **150**, the isolation between the first antenna **130** and the second antenna **150** is improved.

[0026] Furthermore, the grounding element **160** can extend to a part of the inner surface **115** or/and a part of the outer surface **111** corresponding to the speaker element **120**. Thus, adding some metal below the speaker element **120** as part of the grounding element **160** and further enclosing the entire speaker element **120** to enhance grounding can improve the isolation between the first antenna **130** and the second antenna **150**.

[0027] The grounding branch **149** is connected to the grounding element **160** and is closer to the second antenna **150** than the first branch **131** thereto. The first antenna **130** can further include a third branch **143**. One end **144** of the third branch **143** is connected between the one end **132** and

the other end of the first branch **131**. The other end **145** of the third branch **143** can be connected to the grounding element **160** or can be an open end. In this embodiment, the other end **145** of the third branch **143** is specifically connected to the grounding element **160**. Thus, the antenna module **100** according to the present disclosure is applicable when the first antenna is a PIFA antenna, and the type of the second antenna **150** is not restricted. In addition, the specific patterns of the first antenna and the second antenna according to the present disclosure are not limited to those illustrated in the drawings.

[0028] The speaker element **120** has a first speaker edge **127** and a second speaker edge **128** in the third direction z , and the first speaker edge **127** is farther from the second antenna **150** than the second speaker edge **128** therefrom. The first branch **131** extends along the second direction y . The first branch **131** has a first edge **137** and a second edge **138** in the third direction z . The distance between the first edge **137** and the first speaker edge **127** in the third direction z is the first distance $d1$, and the distance between the second edge **138** and the second speaker edge **128** in the third direction z is the second distance $d2$. The first distance $d1$ can be smaller than the second distance $d2$. Accordingly, the first branch **131** and the second antenna **150** are appropriately positioned with a large-area metal surface of the speaker element **120** in between, facilitating a balance between isolation and space utilization efficiency.

[0029] The distance between the first speaker edge **127** and the second speaker edge **128** is the length $m2$ of the speaker element **120**. The ratio of the length $m2$ of the speaker element **120** to the first distance $d1$ can be equal to or greater than 4. Alternatively, in the third direction z , the position of the first branch **131** is set within a range extending from the first speaker edge **127** toward the second speaker edge **128**, up to $\frac{1}{4}$ of the length $m2$ of the speaker element **120**. Thus, isolation can be effectively improved.

[0030] The first antenna **130** and the second antenna **150** have the same operating frequency. The ratio of the length $m2$ of the speaker element **120** to the corresponding wavelength of the same operating frequency can be equal to or greater than $\frac{1}{8}$. Thus, the speaker element **120** can be used as an isolator to effectively improve the isolation between the first antenna **130** and the second antenna **150**. In the first embodiment, as shown in FIG. 1A, the length $m2$ of the speaker element **120** in the third direction z can be 16 mm, and its width in the second direction y can be 8 mm. The first antenna **130** and the second antenna **150** operate at the same operating frequency of 2.4 GHz. The length $m2$ of the speaker element **120**, being 16 mm, corresponds to $\frac{1}{8}$ of the wavelength at 2.4 GHz, making it suitable for use as an isolator.

[0031] Referring to FIG. 1B, in the first direction x , there can be a non-zero gap $s2$ between the speaker element **120** and a part of the inner surface **115** opposite the first antenna **130**. The ratio of the gap $s2$ to the thickness $t2$ of the speaker element **120** is between 2 and 3. Thus, it helps to improve the isolation without affecting the RF characteristics of the first antenna **130**.

[0032] FIG. 1C is an isolation schematic diagram of the antenna module **100** in FIG. 1A. Referring to FIG. 1C, FIG. 1C compares isolations of the antenna module **100** and a prior art antenna module (with two antennas but no metal element), and the isolation is the $S21$ or $S12$ parameter of the two antennas. FIG. 1C shows that the antenna module **100** according to the present disclosure has better isolation (i.e., smaller $S21$ or $S12$ parameter value) at most operating frequencies compared to the prior art antenna module.

[0033] FIG. 2 is a schematic diagram of a speaker module **200** according to a second embodiment of the present disclosure. Referring to FIG. 2, the speaker module **200** of the second embodiment can also be referred to as an antenna module. The speaker module **200** includes a metal element and a first antenna **230**. In specific, the metal element is the speaker element **120**, which includes a metal surface. The first antenna **230** includes a first branch **131**. When observed along the first direction x , the projection of the first branch **131** partially overlaps with the projection of the speaker element **120**.

[0034] Furthermore, the speaker module **200** includes a cavity body **110**, a speaker element **120**,

and a first antenna **230**. The cavity body **110** includes an outer surface **111**, an inner surface **115**, and an accommodation space **116**, and the inner surface **115** forms the accommodation space **116**. The speaker element **120** is disposed in the accommodation space **116**. The first antenna **230** is disposed on and extends along the outer surface **111**.

[0035] The first antenna **230** further includes a third branch **243**. One end **244** of the third branch **243** is connected between one end **132** and the other end of the first branch **131**. The other end **245** of the third branch **243** is specifically an open end.

[0036] The main difference between the speaker module **200** of the second embodiment and the antenna module **100** of the first embodiment is that the other end **245** of the third branch **243** is an open end. Thus, the speaker module **200** according to the present disclosure is applicable when the first antenna **230** is a monopole antenna, and the type of the second antenna **150** is not restricted. Furthermore, other details of the speaker module **200** in the second embodiment can be referenced from the antenna module **100** described in the first embodiment and will not be elaborated herein. It should be noted that the length and width of the branches, as well as the parameter values of the components in the antenna module **100** and the speaker module **200**, may vary depending on requirements.

[0037] FIG. **3** is a schematic diagram of a speaker module **300** according to a third embodiment of the present disclosure. Referring to FIG. **3**, the speaker module **300** of the third embodiment can also be referred to as an antenna module. The speaker module **300** includes a metal element and a first antenna **330**. The metal element is specifically the speaker element **320**, which includes a metal surface. The first antenna **330** includes a first branch **331**. When observed along the first direction x, the projection of the first branch **331** partially overlaps with the projection of the speaker element **320**.

[0038] Moreover, the speaker module **300** includes a cavity body **310**, a speaker element **320**, and a first antenna **330**. The cavity body **310** includes an outer surface **311**, an inner surface **315**, and an accommodation space **316**, and the inner surface **315** forms the accommodation space **316**. The speaker element **320** is disposed in the accommodation space **316**, and the first antenna **330** is disposed on and extends along the outer surface **311**.

[0039] Specifically, the speaker module **300** further includes a grounding element **360**, a second branch **342**, a fourth branch **346**, and a fifth branch **347**. One end **332** of the first branch **331** is connected to the grounding element **360**. When observed along the first direction x, the projection of one end **332** of the first branch **331** does not overlap with the projection of the speaker element **320**. The other end of the first branch **331** is connected to the second branch **342**, and both ends of the second branch **342** are open ends. The fourth branch **346** is farther from the second antenna **350** than the first branch **331** therefrom. One end of the fourth branch **346** is provided with a feeding point **340**, and the other end of the fourth branch **346** is connected to the fifth branch **347**. Both ends of the fifth branch **347** are open ends, and the fifth branch **347** is parallel to, adjacent to, and coupled with the second branch **342**. Thus, the speaker module **300** according to the present disclosure is applicable when the first antenna **330** is a coupled antenna, and the type of the second antenna **350** in the speaker module **300** is not restricted. Furthermore, the fourth branch **346** serves as the feed branch, while the first branch **331** serves as the grounding branch. When observed along the first direction x, the projection of the end of the fourth branch **346** where the feed point **340** is provided does not overlap with the projection of the speaker element **320**, whereas the projection of the first branch **331** partially overlaps with the projection of the speaker element **320**. This arrangement helps balance RF performance requirements and enhances isolation.

[0040] Each of the first antenna **330** and the second antenna **350** extends along the second direction y and the third direction z. The first antenna **330**, the speaker element **320**, and the second antenna **350** are sequentially arranged along the third direction z.

[0041] The cavity body **310** is made of a dielectric material, and each of the first antenna **330** and the second antenna **350** is made of a conductive material. The second antenna **350** is disposed on

and extends along the outer surface **311**, and the grounding element **360** is connected to the outer surface **311**. Furthermore, the grounding element **360** extends to a part of the inner surface **315** or/and a part of the outer surface **311** corresponding to the speaker element **320**.

[0042] The first antenna **330** and the second antenna **350** operate at the same operating frequency. The ratio of the length of the speaker element **320** along the third direction z to the wavelength corresponding to this operating frequency is equal to or greater than $\frac{1}{8}$, and the operating frequency bands of the first antenna **330** and the second antenna **350** may at least partially overlap.

Additionally, in the third direction z , the position of the first branch **331** is set within a range extending from the first speaker edge **327** toward the second speaker edge **328**, up to $\frac{1}{4}$ of the length of the speaker element **320**.

[0043] In the first direction x , there is a non-zero gap between a part of the inner surface **315** opposite the first antenna **330** and the speaker element **320**. The ratio of the gap to the thickness of the speaker element **320** is between 2 and 3.

[0044] FIG. **4** is a schematic diagram of the electronic device **400** according to a fourth embodiment of the present disclosure. Referring to FIG. **1A** and FIG. **4**, the electronic device **400** includes the antenna module or speaker module of the present disclosure, such as the aforementioned antenna module **100**, or the aforementioned speaker modules **200**, **300**. The electronic device **400** is specifically a notebook computer, but the present disclosure is not limited thereto. Thus, it helps to reduce interference between the first antenna **130** and the second antenna **150**, and is suitable for applications in electronic devices **400** with limited space.

[0045] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0046] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A speaker module comprising: a cavity body comprising an outer surface, an inner surface, and an accommodation space, wherein the inner surface forms the accommodation space; a speaker element disposed in the accommodation space; and a first antenna disposed on and extending along the outer surface.
2. The speaker module according to claim 1, wherein the cavity body is made of a dielectric material, and the first antenna is made of a conductive material.
3. The speaker module according to claim 1, wherein when observed along a first direction, a projection of a first branch of the first antenna partially overlaps with a projection of the speaker element.
4. The speaker module according to claim 3, wherein in the first direction, there is a gap between the speaker element and a part of the inner surface opposite the first antenna, and a ratio of the gap to a thickness of the speaker element is between 2 and 3.
5. The speaker module according to claim 3, further comprising: a grounding element connected to the outer surface, wherein one end of the first branch is provided with a feeding point or connected to the grounding element; wherein, when observed along the first direction, a projection of the one end of the first branch does not overlap with the projection of the speaker element.
6. The speaker module according to claim 3, further comprising: a second antenna disposed on and extending along the outer surface; and a grounding element connected to the outer surface; wherein each of the first antenna and the second antenna extends along a second direction and a third direction, the first direction, the second direction, and the third direction are perpendicular to one

another, and the first antenna, the speaker element, and the second antenna are sequentially arranged along the third direction.

7. The speaker module according to claim 6, wherein the speaker element comprises a metal surface, and the grounding element extends to a part of at least one of the inner surface and the outer surface corresponding to the speaker element.

8. The speaker module according to claim 6, wherein an operating frequency band of the first antenna and an operating frequency band of the second antenna at least partially overlap.

9. The speaker module according to claim 6, wherein the speaker element has a first speaker edge and a second speaker edge in the third direction, and the first speaker edge is farther from the second antenna than the second speaker edge therefrom; wherein the first branch extends along the second direction, the first branch has a first edge and a second edge in the third direction, a distance between the first edge and the first speaker edge in the third direction is a first distance, a distance between the second edge and the second speaker edge in the third direction is a second distance, and the first distance is smaller than the second distance.

10. The speaker module according to claim 9, wherein a distance between the first speaker edge and the second speaker edge is a length of the speaker element, and a ratio of the length of the speaker element to the first distance is equal to or greater than 4.

11. The speaker module according to claim 9, wherein a distance between the first speaker edge and the second speaker edge is a length of the speaker element, the first antenna and the second antenna have a same operating frequency, and a ratio of the length of the speaker element to a corresponding wavelength of the same operating frequency is equal to or greater than $\frac{1}{8}$.

12. The speaker module according to claim 6, wherein one end of the first branch is provided with a feeding point, the other end of the first branch is connected to a second branch of the first antenna, and both ends of the second branch are open ends; wherein, when observed along the first direction, a projection of a grounding branch of the first antenna partially overlaps with a projection of the speaker element, and the grounding branch is connected to the grounding element and closer to the second antenna than the first branch thereto; wherein one end of a third branch of the first antenna is connected between the one end and the other end of the first branch, and the other end of the third branch is connected to the grounding element or is an open end.

13. The speaker module according to claim 6, wherein one end of the first branch is connected to the grounding element, the other end of the first branch is connected to a second branch of the first antenna, and both ends of the second branch are open ends; wherein a fourth branch of the first antenna is farther from the second antenna than the first branch therefrom, one end of the fourth branch is provided with a feeding point, the other end of the fourth branch is connected to a fifth branch of the first antenna, both ends of the fifth branch are open ends, and the fifth branch is parallel to and adjacent to the second branch.

14. An electronic device, comprising: the speaker module according to claim 1.

15. An antenna module, comprising: a metal element comprising a metal surface; and a first antenna; wherein, when observed along a first direction, a projection of a first branch of the first antenna partially overlaps with a projection of the metal element.

16. The antenna module according to claim 15, further comprising: a grounding element, wherein one end of the first branch is provided with a feeding point or connected to the grounding element; wherein, when observed along the first direction, a projection of the one end of the first branch does not overlap with the projection of the metal element.

17. The antenna module according to claim 16, wherein the other end of the first branch is connected to a second branch of the first antenna, and both ends of the second branch are open ends.

18. The antenna module according to claim 15, wherein when observed along the first direction, a projection of a grounding branch of the first antenna partially overlaps with the projection of the metal element.

19. The antenna module according to claim 15, further comprising: a second antenna; wherein each of the first antenna and the second antenna extends along a second direction and a third direction, the first direction, the second direction, and the third direction are perpendicular to one another, and the first antenna, the metal element, and the second antenna are sequentially arranged along the third direction.
